

XIAOCHENG YANG

xyang21@wm.edu | Homepage | Google Scholar | GitHub

RESEARCH INTERESTS

LLM Agents · Multi-Agent Systems · AI for Scientist · LLM Reasoning · Natural Language Processing

EDUCATION

College of William & Mary, Williamsburg, VA

Ph.D. in Applied Science (Data Science); Advisor: Prof. Qingyun Wang

Aug 2026 – Present

Expected May 2031

University of Illinois Urbana-Champaign, Urbana, IL

M.S. in Computer Science; Advisors: Prof. Gokhan Tur and Prof. Dilek Hakkani-Tür

Aug 2024 – May 2026

GPA: 3.96/4.00

New York University Shanghai, Shanghai, China

B.S. in Computer Science, Minor in Mathematics; Advisor: Prof. Yik-Cheung (Wilson) Tam

Aug 2020 – May 2024

GPA: 3.92/4.00

RESEARCH EXPERIENCE

Department of Data Science, College of William & Mary

Graduate Research Assistant | Advisor: Prof. Qingyun Wang

Aug 2026 – Present

- Investigating **agentic AI and AI for science**, with a focus on reliable AI systems for scientific knowledge acquisition, reasoning, and discovery.

Siebel School of Computing and Data Science, UIUC

Graduate Research Assistant | Advisors: Prof. Gokhan Tur and Prof. Dilek Hakkani-Tür

Aug 2024 – May 2026

- Developed **Gradient-Based Connections (GBC)**, a credit-assignment method for optimizing prompts in LLM multi-agent systems; released the method in **AgentChord**, an open-source multi-agent framework.
- Designed a prefix-based technique to reduce credit-assignment space complexity to make GBC practically usable; achieved **99.0% Inform**, **94.0% Success**, and **91.4% slot F1** on MultiWOZ and **24.3%** on τ -bench, outperforming strong single-agent baselines.
- Adapted ReAct to task-oriented dialogue and designed friction-aware prompts to reduce interaction ambiguity, improving MultiWOZ Inform and Success by **5.5 and 3.0 points**.
- Built LLM-based question-generation and student-simulation systems for K–2 literacy assessment; achieved **50.4% MAP@1** with ROUGE-L F1 on FairytaleQA, outperforming prior methods.

Department of Computer Science, NYU Shanghai

Undergraduate Research Assistant | Advisor: Prof. Yik-Cheung (Wilson) Tam

Jun 2023 – May 2024

- Fine-tuned LoRA adapters for LLM arithmetic reasoning; compared chain-of-thought, Prolog generation, and hybrid outputs, and open-sourced the **GSM8K-Prolog** dataset.
- Evaluated Prolog-permutation data augmentation, improving accuracy over the chain-of-thought baseline by **10.9 points on GSM8K** and **22.6 points on GSM-HARD**; analyzed divergence between cross-entropy loss and executable-code accuracy.
- Fine-tuned LLMs for dialogue state tracking across model, data, LoRA, and output-format settings, achieving **82.4% joint goal accuracy**, a new state of the art among end-to-end DST methods.

SELECTED PUBLICATIONS

Xiaocheng Yang, Abdulrahman Alrabah, Dilek Hakkani-Tür, Gokhan Tur. *SIGDIAL 2026 (Oral Presentation)*. [Code] “**GBC: Gradient-Based Connections for Optimizing Multi-Agent Systems**”

Xiaocheng Yang, Sumuk Shashidhar, Dilek Hakkani-Tür. *SLaTE 2025*. “**Question Generation for Assessing Early Literacy Reading Comprehension**”

Kunlun Zhu[†], Hongyi Du[†], Zhaochen Hong[†], **Xiaocheng Yang[†]**, Shuyi Guo[†], Zhe Wang[†], et al. *ACL 2025, Main Conference*. [Code] “**MultiAgentBench: Evaluating the Collaboration and Competition of LLM Agents**”

[†] Core contributors; considered co-first authors.

Xiaocheng Yang, Bingsen Chen, Yik-Cheung Tam. *NAACL 2024, Main Conference*. [Code] “**Arithmetic Reasoning with LLM: Prolog Generation & Permutation**”

SELECTED PUBLICATIONS (CONTINUED)

Abdulrahman AlRabah, **Xiaocheng Yang**, Dilek Hakkani-Tür, Abdussalam Alawini. *arXiv preprint*, 2026. “**PredAct-Bench: Benchmarking Tool-Augmented Dialogue under Controlled Tool Noise**”

Jiayu Liu, Qihan Lin, Cheng Qian, Rui Wang, Emre Can Acikgoz, **Xiaocheng Yang**, et al. *arXiv preprint*, 2026. “**PlanBench-XL: Evaluating Long-Horizon Planning of LLM Tool-Use Agents in Large-Scale Tool Ecosystems**”

Shuhaib Mehri, **Xiaocheng Yang**, Takyoun Kim, et al. *Transactions of the Association for Computational Linguistics (TACL)*, 2026. “**Goal Alignment in LLM-Based User Simulators for Conversational AI**”

Nimet Beyza Bozdog, Shuhaib Mehri, **Xiaocheng Yang**, et al. *ACM Computing Surveys*, 2026. “**Must Read: A Comprehensive Survey of Computational Persuasion**”

Cheng Qian, Peixuan Han, Qinyu Luo, . . . , **Xiaocheng Yang**, et al. *ACL 2025, Main Conference*. “**EscapeBench: Towards Advancing Creative Intelligence of Language Model Agents**”

Vardhan Dongre, **Xiaocheng Yang**, Emre Can Acikgoz, et al. *IWSDS 2025*. “**ReSpAct: Harmonizing Reasoning, Speaking, and Acting Towards Building LLM-Based Conversational AI Agents**”

TEACHING EXPERIENCE

College of William & Mary	Williamsburg, VA
<i>Teaching Assistant, DATA 101: Reasoning with Generative AI</i>	Fall 2026
<i>Student Success Tutor</i>	Fall 2026

ACADEMIC SERVICE

Reviewer: NeurIPS 2026; ACL Rolling Review, 2025–2026 (3 cycles)

HONORS & AWARDS

Summa Cum Laude; Honors in Computer Science; NYU Shanghai Excellence Award; Dean's List (2020–2023)

TECHNICAL SKILLS

Programming: Python, SQL, Java, R
ML / LLM: PyTorch, Hugging Face Transformers, PEFT, DeepSpeed, torchrun, scikit-learn
Data & Tools: Pandas, NumPy, Matplotlib, Git, Linux, LaTeX